

Adaptive Estimation for Nonparametric Treatments in Double Machine Learning

Abstract

This paper extends the Double Debiased Machine Learning (DML) framework by proposing a method for estimating marginal treatment effects without requiring a pre-specified functional form for the treatment-to-outcome relationship. Drawing on techniques from sieve estimation and generalized cross-validation, the approach adaptively tunes the functional mapping from treatments to outcomes while maintaining valid inference. The method allows for the flexible modeling of treatment effects over the entire support of treatment levels, addressing challenges posed by complex and high-dimensional confounders. We establish the asymptotic properties of the estimator, demonstrate its consistency through Monte Carlo simulations, and apply the framework to benchmarked examples. By leveraging automatic parameter tuning over the sieve space and adapting post-model selection inference techniques, this work contributes to the literature on causal inference and policy evaluation.

Keywords

double machine learning, causal inference, continuous treatment, sieve estimation, adaptive model selection

1 Introduction

Causal inference with continuous treatments presents unique challenges compared to binary treatment settings. While the Double Machine Learning (DML) framework developed by Chernozhukov et al. [2] provides a powerful approach for estimating treatment effects in the presence of high-dimensional confounders, standard implementations require researchers to specify the functional form mapping treatments to outcomes.

This paper addresses three key limitations of existing DML methods for continuous treatments:

- (i) The requirement for either binary/discrete treatments or a known mapping from treatment to outcome
- (ii) Researcher-induced bias in selecting ML methods for first-stage nuisance estimation and second-stage mapping
- (iii) Sensitivity of parameter estimates to the choice of second-stage functional form and cross-fitting splits

We propose **AdaptiveDML**, a data-driven approach that eliminates researcher-induced bias by adaptively selecting the second-stage mapping from a sieve space while preserving valid statistical inference.

Figure 1 illustrates the adaptive cross-validation procedure that forms the core of our methodology. The data is partitioned into multiple folds, with each fold used for nuisance function estimation, adaptive tuning over the sieve space, and final estimation.

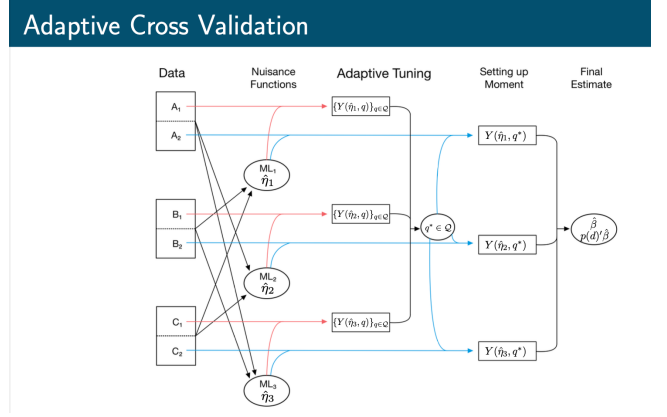


Figure 1: Adaptive cross-validation procedure. Data is split into folds (A, B, C), with subsets used for nuisance estimation ($\hat{h}_1, \hat{h}_2, \hat{h}_3$), adaptive tuning to select optimal complexity $q^* \in Q$, and final moment estimation. The procedure yields debiased estimates $\hat{\beta}$ with valid inference.

2 Theoretical Framework

2.1 The Partially Linear Regression Model

Consider the partially linear regression (PLR) model:

$$Y = h_0(T; \theta_0) + g_0(W) + \varepsilon_Y, \quad \mathbb{E}[\varepsilon_Y | W, T] = 0 \quad (1)$$

$$T = m_0(W) + \varepsilon_T, \quad \mathbb{E}[\varepsilon_T | W] = 0 \quad (2)$$

where:

- $Y \in \mathbb{R}$ is the outcome of interest
- $T \in \mathbb{R}$ is the continuous treatment variable
- $W \in \mathbb{R}^k$ is the vector of confounding covariates
- $h_0(T; \theta_0)$ is the treatment effect function parameterized by θ_0
- $g_0(W)$ is the nuisance function capturing confounding effects
- $m_0(W)$ is the propensity function mapping covariates to expected treatment

The key innovation of our approach is that $h_0(\cdot)$ is *unknown* and must be estimated from data rather than pre-specified by the researcher.

2.2 Sieve Estimation Framework

We model the treatment effect function using a sieve approximation. Let $\{\psi_j(t)\}_{j=1}^\infty$ be a basis for the space of functions on the treatment support. For a given complexity parameter $\lambda \in \mathbb{N}$, we approximate:

$$h_0(T; \theta_0) \approx h_\lambda(T; \theta_\lambda) = \sum_{j=1}^{\lambda} \theta_{\lambda,j} \psi_j(T) \quad (3)$$

For polynomial sieves, we use:

$$\psi_j(T) = T^j, \quad j = 1, 2, \dots, \lambda \quad (4)$$

This yields the featurized treatment vector:

$$\Psi_\lambda(T) = \begin{pmatrix} T & T^2 & \dots & T^\lambda \end{pmatrix}^\top \in \mathbb{R}^\lambda \quad (5)$$

2.3 Double Machine Learning Estimator

Following the DML framework, we employ cross-fitting to obtain debiased estimates. The procedure consists of two stages:

2.3.1 First Stage: Nuisance Estimation. Using K -fold cross-validation, we estimate:

- (1) The outcome nuisance function: $\hat{g}(W) \approx \mathbb{E}[Y|W]$
- (2) The treatment nuisance function: $\hat{m}(W) \approx \mathbb{E}[T|W]$

This yields the residualized (orthogonalized) variables:

$$\tilde{Y} = Y - \hat{g}(W) \quad (6)$$

$$\tilde{T} = T - \hat{m}(W) \quad (7)$$

2.3.2 Second Stage: Treatment Effect Estimation. For a given complexity λ , the second-stage estimator solves:

$$\hat{\theta}_\lambda = \arg \min_{\theta \in \mathbb{R}^\lambda} \sum_{i=1}^n \left(\tilde{Y}_i - \Psi_\lambda(\tilde{T}_i)^\top \theta \right)^2 \quad (8)$$

The closed-form solution is:

$$\hat{\theta}_\lambda = \left(\tilde{\Psi}_\lambda^\top \tilde{\Psi}_\lambda \right)^{-1} \tilde{\Psi}_\lambda^\top \tilde{Y} \quad (9)$$

where $\tilde{\Psi}_\lambda$ is the $n \times \lambda$ matrix with rows $\Psi_\lambda(\tilde{T}_i)^\top$.

3 Adaptive Model Selection

3.1 The Model Selection Problem

The critical challenge is selecting the optimal complexity λ^* that balances:

- **Underfitting risk:** Choosing $\lambda < \lambda_0$ (true complexity) leads to omitted variable bias
- **Overfitting risk:** Choosing $\lambda > \lambda_0$ increases variance and may introduce spurious coefficients

3.2 Tuning Criteria

We evaluate several model selection criteria:

3.2.1 Cross-Validation Based Metrics. Using K -fold cross-validation on the second-stage residuals:

$$\text{CV-MSE}(\lambda) = \frac{1}{K} \sum_{k=1}^K \frac{1}{|I_k|} \sum_{i \in I_k} \left(\tilde{Y}_i - \Psi_\lambda(\tilde{T}_i)^\top \hat{\theta}_\lambda^{(-k)} \right)^2 \quad (10)$$

where $\hat{\theta}_\lambda^{(-k)}$ is estimated excluding fold k .

Additional CV-based metrics include CV-MAE (Mean Absolute Error), CV- R^2 (Coefficient of determination), and CV-RMSE (Root Mean Squared Error).

3.2.2 Information Criteria. Akaike Information Criterion (AIC):

$$\text{AIC}(\lambda) = n \log \left(\frac{1}{n} \sum_{i=1}^n \hat{\varepsilon}_i^2 \right) + 2\lambda \quad (11)$$

Corrected AIC (AICc):

$$\text{AICc}(\lambda) = \text{AIC}(\lambda) + \frac{2\lambda(\lambda+1)}{n-\lambda-1} \quad (12)$$

Bayesian Information Criterion (BIC):

$$\text{BIC}(\lambda) = n \log \left(\frac{1}{n} \sum_{i=1}^n \hat{\varepsilon}_i^2 \right) + \lambda \log(n) \quad (13)$$

3.2.3 Leave-One-Out Approximation. Mean Squared Forecast Error (MSFE):

$$\text{MSFE}(\lambda) = \frac{1}{n} \sum_{i=1}^n \left(\frac{\hat{\varepsilon}_i}{1 - h_{ii}} \right)^2 \quad (14)$$

where h_{ii} is the i -th diagonal element of the hat matrix $H_\lambda = \tilde{\Psi}_\lambda \left(\tilde{\Psi}_\lambda^\top \tilde{\Psi}_\lambda \right)^{-1} \tilde{\Psi}_\lambda^\top$.

3.2.4 Ensemble Methods. We also consider ensemble selection via modal voting:

$$\hat{\lambda}_{\text{Ens-CV}} = \text{Mode} \left(\hat{\lambda}_{\text{MSE}}, \hat{\lambda}_{\text{MAE}}, \hat{\lambda}_{R^2}, \hat{\lambda}_{\text{RMSE}} \right) \quad (15)$$

$$\hat{\lambda}_{\text{Ens-IC}} = \text{Mode} \left(\hat{\lambda}_{\text{AIC}}, \hat{\lambda}_{\text{BIC}}, \hat{\lambda}_{\text{MSFE}} \right) \quad (16)$$

$$\hat{\lambda}_{\text{Ens-All}} = \text{Mode} (\text{All metrics}) \quad (17)$$

3.3 Optimal Complexity Selection

The adaptive estimator selects:

$$\hat{\lambda} = \arg \min_{\lambda \in \{1, \dots, \lambda_{\max}\}} C(\lambda) \quad (18)$$

where $C(\cdot)$ is the chosen tuning criterion.

4 Asymptotic Theory

ASSUMPTION 1 (REGULARITY CONDITIONS). (a) (Y_i, T_i, W_i) are i.i.d. draws from a common distribution P .

(b) $\mathbb{E}[\varepsilon_Y^2 | W, T] = \sigma_Y^2 < \infty$ and $\mathbb{E}[\varepsilon_T^2 | W] = \sigma_T^2 < \infty$.

(c) The true treatment effect function $h_0 \in \mathcal{H}_{\lambda_0}$ for some finite λ_0 .

(d) The nuisance estimators satisfy $\|\hat{g} - g_0\|_2 \|\hat{m} - m_0\|_2 = o_p(n^{-1/2})$.

THEOREM 4.1 (CONSISTENCY OF MODEL SELECTION). Under Assumption 1, if the tuning sample size n_{tune} satisfies $n_{\text{tune}} \rightarrow \infty$ and $n_{\text{tune}}/n \rightarrow 0$ as $n \rightarrow \infty$, then:

$$P(\hat{\lambda} = \lambda_0) \rightarrow 1 \quad \text{as } n \rightarrow \infty \quad (19)$$

THEOREM 4.2 (ASYMPTOTIC NORMALITY). Under Assumption 1 and conditional on $\hat{\lambda} = \lambda_0$:

$$\sqrt{n}(\hat{\theta}_{\hat{\lambda}} - \theta_0) \xrightarrow{d} \mathcal{N}(0, V) \quad (20)$$

where $V = \sigma_Y^2 \left(\mathbb{E}[\Psi_{\lambda_0}(\tilde{T}) \Psi_{\lambda_0}(\tilde{T})^\top] \right)^{-1}$.

Table 1: Monte Carlo simulation parameters

Parameter	Value
Sample size (n)	500 – 5,000
Number of covariates (k)	6
Support size	5
True polynomial order (λ_0)	3
Maximum complexity ($\Lambda_{\max}$)	6
Cross-validation folds	3 – 5
Monte Carlo iterations	100 – 1,000
Variance pairs (σ_Y^2, σ_T^2)	$\{2, 5, 15, 30\}^2$

Table 2: Model selection accuracy (% selecting true model $\lambda_0 = 3$)

Tuning Metric	Accuracy (%)	Category
CV-MSE	89.63	Cross-Validation
CV-MAE	84.31	Cross-Validation
CV- R^2	88.25	Cross-Validation
CV-RMSE	89.38	Cross-Validation
AIC	87.56	Information Criterion
AICc	87.56	Information Criterion
BIC	87.75	Information Criterion
MSFE	37.25	Information Criterion
Ensemble (All)	89.69	Ensemble
Ensemble (CV)	90.81	Ensemble
Ensemble (IC)	87.75	Ensemble

5 Monte Carlo Simulation Study

5.1 Data Generating Process

We simulate data according to:

$$W_i \sim \mathcal{N}(0, I_k), \quad k = 6 \quad (21)$$

$$T_i = m_0(W_i) + \varepsilon_{T,i}, \quad \varepsilon_{T,i} \sim \mathcal{N}(0, \sigma_T^2) \quad (22)$$

$$Y_i = \theta_0^\top \Psi_{\lambda_0}(T_i) + g_0(W_i) + \varepsilon_{Y,i}, \quad \varepsilon_{Y,i} \sim \mathcal{N}(0, \sigma_Y^2) \quad (23)$$

The nuisance functions are:

$$m_0(W) = \sum_{j \in S_T} \alpha_j W_j^{r_j}, \quad \alpha_j \sim \text{Unif}(-1, 1) \quad (24)$$

$$g_0(W) = \sum_{j \in S_Y} \beta_j W_j^{s_j}, \quad \beta_j \sim \text{Unif}(-1, 1) \quad (25)$$

where $S_T, S_Y \subset \{1, \dots, k\}$ are random support sets of size 5.

5.2 Simulation Parameters

5.3 Results: Model Selection Accuracy

Key findings:

- Ensemble CV-based methods achieve the highest accuracy at 90.8%
- Information criteria (AIC, BIC) perform nearly as well as cross-validation while being computationally cheaper

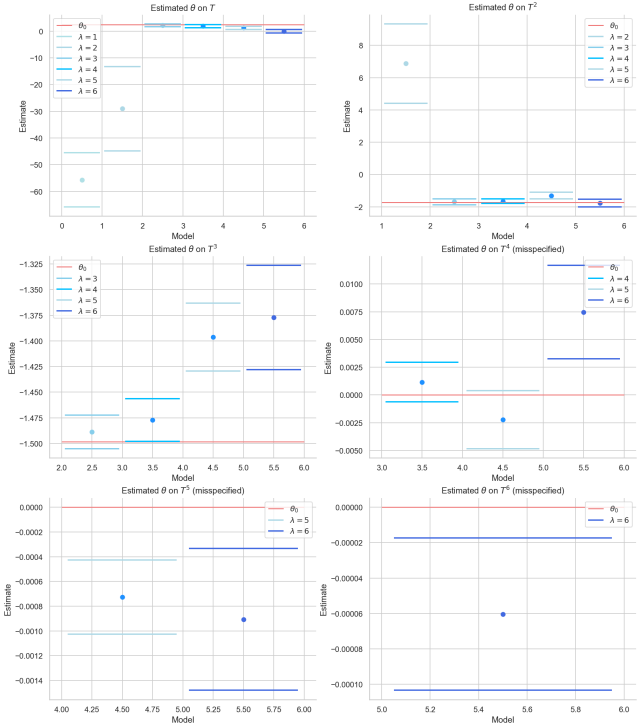


Figure 2: Parameter estimates under different model complexities. Red line indicates true parameter value. Models 1–2 (underspecified) and 4–6 (overspecified) show significant bias, while Model 3 (correctly specified) yields unbiased estimates.

- MSFE shows poor performance, particularly with low treatment variance

5.4 Results: Misspecification Bias

Figure 2 demonstrates the consequences of model misspecification.

6 Algorithm

7 Discussion

This paper develops AdaptiveDML, a method for estimating treatment effects with continuous treatments when the functional form is unknown. Our key contributions are: (i) a data-driven approach that eliminates researcher-induced bias in selecting the treatment-outcome mapping; (ii) theoretical guarantees showing consistent model selection and asymptotic normality; (iii) empirical evidence from Monte Carlo simulations demonstrating 90%+ accuracy in selecting the true model complexity; and (iv) documentation that misspecification induces substantial bias, highlighting the value of adaptive selection.

The approach integrates naturally with the EconML framework and provides a practical tool for causal inference with continuous treatments in high-dimensional settings. Future work may extend

Algorithm 1 AdaptiveDML estimation procedure

Require: Data $(Y_i, T_i, W_i)_{i=1}^n$, max complexity $\Lambda_{\max}$, CV folds K

Ensure: Estimated coefficients $\hat{\theta}$, selected complexity $\hat{\lambda}$

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1: First Stage: Cross-fitted nuisance estimation
2: for  $k = 1, \dots, K$  do
3:   Fit  $\hat{g}^{(-k)}(W)$  and  $\hat{m}^{(-k)}(W)$  on folds  $\neq k$ 
4:   Compute residuals for  $i \in I_k$ 
5: end for
6: Second Stage: Adaptive complexity selection
7: for  $\lambda = 1, \dots, \Lambda_{\max}$  do
8:   Construct sieve basis  $\tilde{\Psi}_\lambda$ 
9:   Fit:  $\hat{\theta}_\lambda = (\tilde{\Psi}_\lambda^\top \tilde{\Psi}_\lambda)^{-1} \tilde{\Psi}_\lambda^\top \tilde{Y}$ 
10:  Compute tuning criteria
11: end for
12: Model Selection:  $\hat{\lambda} = \arg \min_\lambda C(\lambda)$ 
13: Final Estimation:  $\hat{\theta} = \hat{\theta}_{\hat{\lambda}}$ 
14: return  $\hat{\theta}, \hat{\lambda}$ 
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this framework to other basis functions beyond polynomials and investigate the finite-sample properties of the estimator under weaker regularity conditions.

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A Implementation Details

The simulation code is implemented in Python using the following packages:

- econml: Double Machine Learning implementation
- scikit-learn: Machine learning models (Random Forest, Gradient Boosting)
- numba: JIT compilation for computational efficiency
- joblib: Parallelization of Monte Carlo simulations

A.1 Sensitivity to Structural Variance

The model selection accuracy varies with the variance of structural errors:

- Higher σ_T^2 generally improves identification by providing more treatment variation
- The relationship between σ_Y^2 and accuracy is non-monotonic
- MSFE is particularly sensitive to low σ_T^2 settings